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BAULKHAM HILLS HIGH SCHOOL

YEAR 12

YEAR 12 HALF YEARLY EXAMINATION

2006

MATHEMATICS EXTENSION 1

*Time allowed - Two hours
(Plus five minutes reading time)*

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- Start each of the 7 questions on a new page.
- All necessary working should be shown. All marks shown are a guide only
- Write your teacher's name and your name on the cover sheet provided.
- At the end of the exam, staple your answers in order behind the cover sheet provided.

QUESTION 1 (START A NEW PAGE)

(a) Solve $\frac{x}{x-3} \geq 2$

(b) Differentiate

i) $\tan(e^{3x})$

ii) $\ln\left(\frac{2x+1}{x^2}\right)$

(c) Find the acute angle between the lines $y = 2x - 1$ and $x + 3y = 6$

(d) Use Simpson's rule once to estimate

$$\int_2^1 \ln x^2 dx$$

QUESTION 2 (START A NEW PAGE)

(a) Find the Cartesian equation for

$$x = 5 \cos t$$

$$y = 5 \sin t$$

(b) Given that $\sin \alpha = \frac{7}{4}$ and $\sin \beta = \frac{13}{5}$, α, β are acute, find the exact value of $\cos(\alpha + \beta)$

(c) Find the equation of the normal to the curve $y = 2 \log_e x$ at the point (e, 2)

(d) (i) Show that $\frac{1 + \cos 2A}{\sin 2A} = \cot A$

(ii) Hence find the exact value of $\cot 15^\circ$

(e) Find the volume generated by $y = 2 + \cos x$ around the x-axis between

$$x = \frac{4}{\pi} \text{ and } \frac{2}{\pi}$$

QUESTION 3 (START A NEW PAGE)

- (a) Prove by mathematical induction that $1 \times 2^0 + 2 \times 2^1 + 3 \times 2^2 + \dots + n \times 2^{n-1} = 1 + (n-1) \times 2^n$ for all integers $n \geq 1$

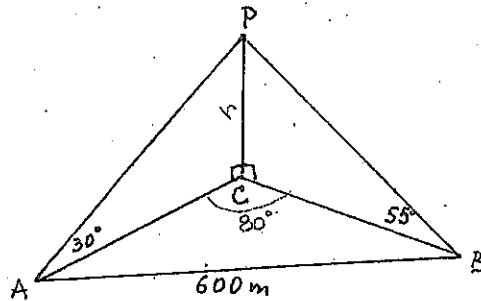
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- (b) Differentiate $y = e^{x^3}$ and hence evaluate

3

$$\int_{-1}^1 x^2 e^{x^3} dx$$

(c)



NOT TO SCALE

Two yachts A and B subtend an angle of 80° at the base C of a cliff. From yacht A the angle of elevation of the point P, h metres vertically above C, is 30° . From yacht B the angle of elevation of the point P is 55° . Yacht B is 600 metres from A.

- (i) Show that $h^2 = \frac{600^2}{\cot^2 30^\circ + \cot^2 55^\circ - 2 \cot 30^\circ \cot 55^\circ \cos 80^\circ}$

3

- (ii) Find h .

1

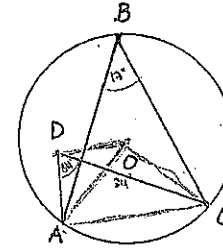
QUESTION 4 (START A NEW PAGE)

- (a) $T(2t, t^2)$ is a point on the parabola $x^2 = 4y$ with focus S. P is the point which divides ST internally in the ratio 1:2.

- (i) What are the coordinates of the focus S? 1
 (ii) Write down the coordinates of P in terms of t 2
 (iii) Hence show that as T moves on the parabola $x^2 = 4y$, the locus of P is the parabola $9x^2 = 12y - 8$. 2

QUESTION 4 continued

(b)



$$\angle AOC = 34^\circ$$

$$\angle ADC = 34^\circ$$

$\therefore$ ADOC lie on a circumference
 $\therefore$ $\angle$'s subtended on equal arcs to circumference
 $\therefore$ ADOC must be cyclic
 NOT TO SCALE

Points A, B and C lie on the circumference of a circle with centre O. Point D lies inside the circle and $\angle ABC = 17^\circ$ and $\angle ADC = 34^\circ$. Prove that ADOC is a cyclic quadrilateral.

3

- (c) Solve $3 \sin \theta - 4 \cos \theta = 3$ for $0^\circ \leq \theta \leq 360^\circ$.

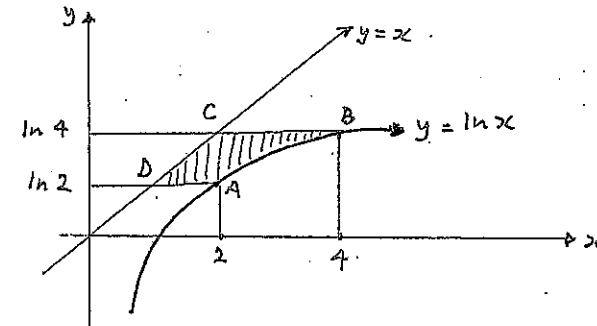
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QUESTION 5 (START A NEW PAGE)

- (a) A function is defined by the rule $F(x) = 2x.e^x$

- (i) Find the stationary point and determine its nature. 4
 (ii) Sketch the graph $F(x)$ 2

- (b) The diagram below shows the graphs of $y = x$ and $y = \ln x$. A(2, ln 2) and B(4, ln 4) are points on $y = \ln x$. C and D are points on $y = x$. Find the shaded area ABCD. Assume BC and AD are parallel to the x-axis. 3



NOT TO SCALE

- (c) The rate of flow of water into, then out of a container is given by $R = t(10 - t)$ litres/minute.

- (i) Find an expression for the volume, V litres, of water in the container at time t minutes assuming that the container is initially empty. 2
 (ii) Find the total time for the container to fill and then empty. 2

QUESTION 6 (START A NEW PAGE)

- (a) The velocity
- v
- m/s of an object at time
- t
- seconds is given by

$$v = 3t^2 - 14t + 8.$$

The object is initially 30 metres to the right of the origin.

- (i) Find the initial acceleration of the particle
 (ii) Find when the object is at rest
 (iii) Find the minimum distance between the origin and the object during its motion

2
2
3

- (b) The rate at which a body cools in air is proportional to the difference between the temperature, T , of the body and the constant surrounding temperature S . This can be expressed as

$$\frac{dT}{dt} = k(T - S)$$

where t is time in minutes and k is constant

- (i) Show that $T = S + Be^{kt}$, where B is a constant is a solution of the above equation.
 (ii) If a particular body cools from 100° Celsius to 80° Celsius in 30 minutes, find the temperature of the body after a further 30 minutes, given that the surrounding temperature remains constant at 25° Celsius.

1
3

QUESTION 7 (START A NEW PAGE)

- (a) (i) State the largest positive domain for which $f(x) = x^2 - 2x + 3$ has an inverse function and sketch the curve for this domain.
 (ii) Find the inverse function $f^{-1}(x)$ of $f(x) = x^2 - 2x + 3$ and state its domain
 (iii) On the same set of axes draw a neat sketch of the inverse function
 (iv) Evaluate $f(f^{-1}(8))$

2
3
1
1

QUESTION 7 continued

- (b) A man borrows \$15800 with monthly reducible interest of 8% p.a. Let A_n be the amount owing at the end of n months, after the monthly repayment has been made.
 (i) If the repayments are \$ M per month, show that after the second repayment he still owes $A_2 = 15800 \times \left(\frac{151}{150}\right)^2 - M\left(\frac{151}{150} + 1\right)$
 (ii) Show that after n repayments the amount owing is $A_n = 15800 \times \left(\frac{151}{150}\right)^n - M\left[\left(\frac{151}{150}\right)^{n-1} + \left(\frac{151}{150}\right)^{n-2} + \dots + 1\right]$
 (iii) If the repayments are \$1260 per month, find the number of payments to repay all the loan.

2
1
3

TOTAL (84)

Question 1 (12)

a) $\frac{x}{x-3} > 2 \quad | \cdot (x-3)^2$ (3)
 $x(x-3) > 2(x-3)^2$
 $0 > (x-3)(x-6)$ (1)
 $3 < x < 6$ (1)

b) i) $\frac{d}{dx} \tan(e^{3x}) = \sec^2(e^{3x}) \cdot 3e^{3x}$ (1)
 ii) $\frac{d}{dx} \ln\left(\frac{2x+1}{x^2}\right) = \frac{2}{2x+1} - \frac{2}{x}$ (1)
 $= \frac{-2x^2+2x}{(2x+1) \cdot x^2}$ (1)

c) $y = 2x - 1$ $2x + 3y = 6$
 $m_1 = 2$ $m_2 = -\frac{1}{3}$
 $\tan \theta = \frac{|m_1 - m_2|}{|1 + m_1 m_2|} = \frac{|2 + \frac{1}{3}|}{|1 - \frac{2}{3}|} = 7$ (1)
 $\therefore \theta = 81^\circ 52'$ (1)

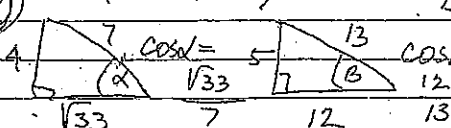
d) $\int \ln x^2 dx$

x	1	1.5	2
$\ln x^2$	0	0.8109	1.3863

 (1)
 $\therefore \int \ln x^2 dx = \frac{0.5}{3} (0 + 4 \times 0.8109 + 1.3863)$ (1)
 $= 0.77165$ (1)

Question 2 (13)

a) $x = 5 \cos t$ $y = 5 \sin t$ (2)
 $x^2 + y^2 = 5^2 (\sin^2 t + \cos^2 t)$ (1)
 $\therefore x^2 + y^2 = 25$ (1)

b) Given $\sin \alpha = \frac{4}{7}$ $\sin \beta = \frac{5}{13}$ (2)

 $\therefore \cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$
 $= \frac{\sqrt{33}}{7} \cdot \frac{12}{13} - \frac{4}{7} \cdot \frac{5}{13}$
 $= \frac{12\sqrt{33} - 20}{91}$ (1)

c) $y = 2 \log_e x$ at $(e, 2)$ (2)
 $\frac{dy}{dx} = \frac{2}{x} \therefore m = \frac{2}{e}$ (1)
 $\therefore$ normal $y - 2 = -\frac{e}{2}(x - e)$ (1)

d) LHS = $\frac{1 + \cos 2A}{\sin 2A} = \frac{1 + 2\cos^2 A - 1}{2\sin A \cos A}$ (1)
 $= \frac{2\cos^2 A}{2\sin A \cos A} = \frac{\cos A}{\sin A} = \cot A$ (1)
 $=$ RHS $\therefore$ proven (3)
 ii) $\cot 15^\circ = \frac{1 + \cos 30^\circ}{\sin 30^\circ} = \frac{1 + \frac{\sqrt{3}}{2}}{\frac{1}{2}}$ (1)

Question 2 - cont.

e) $y = 2 + \cos x$ (4)
 $V = \pi \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} (2 + \cos x)^2 dx$
 $= \pi \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} 4 + 4\cos x + \cos^2 x dx$ (1)
 $= \pi \left[4x + 4\sin x + \frac{1}{2} \cos 2x + \frac{1}{2} x \right]_{\frac{\pi}{4}}^{\frac{3\pi}{4}}$ (1)
 $= \pi \left[2\pi + 4 + 0 + \frac{\pi}{4} - \frac{4}{\sqrt{2}} - \frac{1}{2} - \frac{\pi}{8} \right]$ (1)
 $= \pi \left(\frac{9\pi}{8} - 2\sqrt{2} + 3\frac{3}{4} \right)$ (1)

Question 3 (11)

a) STEP 1 - prove true for $n=1$ (4)
 LHS = $1 \times 2 = 1$
 RHS = $1 + (1-1) \times 2 = 1$ $\therefore$ true (1)
 STEP 2 $\therefore$ assume true for $n=k$
 $\therefore 1 \times 2^0 + 2 \times 2^1 + 3 \times 2^2 + \dots + k(2^{k-1}) = 1 + (k-1) \cdot 2^k$ (1)
 then prove for $n=k+1$
 $\therefore 1 \times 2^0 + (k+1)2^1 = 1 + (k+1) \cdot 2^{k+1}$ (1)
 $= 1 + k \cdot 2^{k+1}$ (1)

Proof
 LHS = $1 \times 2^0 + 2 \times 2^1 + \dots + k \times 2^{k-1} + (k+1) \cdot 2^k$
 $=$ assumption

$= 1 + (k-1) \cdot 2^k + (k+1) \cdot 2^{k+1}$ (1)
 $= 1 + k \cdot 2^k - 2^k + k \cdot 2^{k+1} + 2^k$
 $= 1 + 2 \cdot k \cdot 2^k = 1 + k \cdot 2^{k+1}$
 $=$ RHS $\therefore$ proven
 Proven true for $n=1$, if true for $n=k$ and proven true for $n=k+1$, since proven true for $n=1 \therefore$ true for $n=2, 3, \dots$
 $\therefore$ true for all $n \geq 1$.

b) $y = e^{x^3} \therefore \frac{dy}{dx} = 3x^2 \cdot e^{x^3}$ (1)
 $\therefore \int x^2 \cdot e^{x^3} dx = \frac{1}{3} y = \frac{1}{3} e^{x^3}$ (1)
 $= \frac{1}{3} e - \frac{1}{3} e^{-1}$ (1)

c) i) $\cot 30^\circ = \frac{AC}{h} \therefore AC = h \cdot \cot 30^\circ$ (1)
 ii) $\cot 55^\circ = \frac{BC}{h} \therefore BC = h \cdot \cot 55^\circ$ (1)
 and $600^2 = AC^2 + BC^2 - 2 \cdot AC \cdot BC \cdot \cos 80^\circ$ (1)
 $600^2 = h^2 \cot^2 30^\circ + h^2 \cot^2 55^\circ - 2 \cdot h \cot 30^\circ \cot 55^\circ$ (1)
 $\therefore h^2 = \frac{600^2}{\cot^2 30^\circ + \cot^2 55^\circ - 2 \cot 30^\circ \cot 55^\circ}$ (1)

ii) $h = 342.489 \text{ m}$ (1)

Question 4 (11)

2) $T(2t, t^2)$ $x^2 = 4y$

1) $x^2 = 4ay \therefore a=1$

$\therefore S(0,1)$ ①

1) $S(0,1)$ $T(2t, t^2)$

1 : 2

$P\left(\frac{2t}{3}, \frac{2t^2}{3}\right)$

① ①

Locus of P

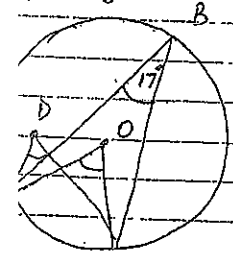
$x = \frac{2t}{3}$

$y = \frac{2t^2}{3}$

$t = \frac{3x}{2}$ $y = \frac{2 + \left(\frac{3x}{2}\right)^2}{3}$ ①

$3y = 2 + \frac{9x^2}{4}$

$12y = 8 + 9x^2 \therefore$ shown



C

CE $\angle ABC = 17^\circ \therefore \angle AOC = 34^\circ$

angle at the centre is double the angle at the circumf. ①

But $\angle AOC = 34^\circ$ and both

$\angle AOC$ and AOC are standing on the same arc and they are equal... AOC is cyclic

c) Solve $3\sin\theta - 4\cos\theta = 3$

$0^\circ \leq \theta \leq 360^\circ$

$A = \sqrt{3^2 + 4^2} = 5$ ①

$3\sin\theta - 4\cos\theta = A\sin(\theta - \alpha)$

$\alpha \therefore \tan\alpha = \frac{4}{3} \therefore \alpha = 53.8^\circ$ ①

$\therefore 5\sin(\theta - 53.8^\circ) = 3$

$\sin(\theta - 53.8^\circ) = \frac{3}{5}$

$\theta - 53.8^\circ = 36.8^\circ$ $\theta = 53.8^\circ + 36.8^\circ = 90.6^\circ$

$\theta = 90^\circ$ or $\theta = 196.76^\circ$

① ①

Question 5 (13)

a) $F(x) = 2x \cdot e^x$

i) $F'(x) = 2e^x + 2xe^x$ ①

$0 = 2e^x(1+x)$

$2e^x \neq 0$ at $x = -1$ ①

$\therefore y = \frac{-2}{e}$ ① st. point

[nature] $F''(x) = 2e^x + 2e^x + 2xe^x$

$F''(x) = 4e^x + 2xe^x$

$F''(x) \text{ at } x = -1 = 0.74 > 0 \therefore \text{min}$ ①

iii) Sketch: x -int: $(0,0)$

$x \rightarrow \infty \therefore e^x \rightarrow 0 \therefore y = 0$ asympt

ii) pt. of infl: $F''(x) = 0$

$4e^x + 2xe^x = 0$

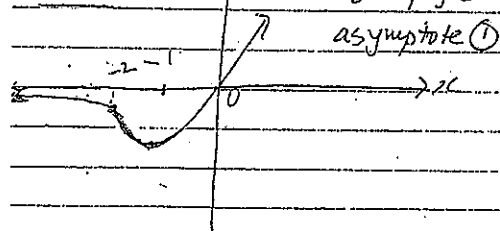
$2e^x(2+x) = 0$

$x = -2$

pt. of infl $(-2, \frac{-4}{e^2})$

iii)

Sketch



b) $y = \ln x \therefore e^y = x$

$A_{\text{shaded}} = \int_{\ln 2}^{\ln 4} e^y dy - \int_{\ln 2}^{\ln 4} y dy$ ①

$= \left[e^y \right]_{\ln 2}^{\ln 4} - \left[\frac{y^2}{2} \right]_{\ln 2}^{\ln 4}$ ①

$= (4-2) - (0.72) = 1.28$ (2dp)

or in exact $A = 2 - \frac{1}{2} \times \ln 2 \times \ln 8$

or $A = 2 - \frac{3}{2} (\ln 2)^2$

(by using $A = \int_{\ln 2}^{\ln 4} e^y dy - A_{\text{TRAPAZIUM ADCB}}$)

c) $R = \frac{1}{2}(10-t)$ in litres/min

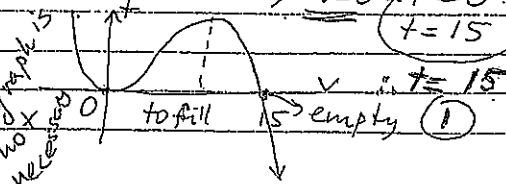
i) $R = \frac{dV}{dt} \therefore V = \int \frac{1}{2}(10-t) dt$

$V = \int 10t - \frac{1}{2}t^2 dt = 5t^2 - \frac{t^3}{3} + C$ ①

when $t=0 \therefore V=0 \therefore C=0$ ①

$\therefore V = 5t^2 - \frac{t^3}{3}$

ii) $V = t^2(5 - \frac{t}{3})$ $V=0 \therefore t=0$ $t=15$ ①



Question 6

11

$$v = 3t^2 - 14t + 8$$

$$t = 0, x = +30m$$

$$a = v = 6t - 14 \quad (1)$$

$$a(t=0) = -14m/s^2 \quad (1)$$

$$\text{rest } \therefore v = 0 = 3t^2 - 14t + 8 \quad (1)$$

$$0 = (3t-2)(t-4)$$

$$t = \frac{2}{3}, t = 4 \quad (1)$$

$$i) \text{ min. distance } \therefore v = 0$$

$$t = \frac{2}{3}, t = 4$$

$$x = \int v dt = \int 3t^2 - 14t + 8 dt$$

$$x = t^3 - 7t^2 + 8t + C$$

$$30 = 0 - 0 + 0 + C \therefore C = 30$$

$$x = t^3 - 7t^2 + 8t + 30 \quad (1)$$

$$x(t = \frac{2}{3}) = 35 \frac{23}{27}$$

$$x(t = 4) = 14$$

$$\therefore \text{min. distance is } 14m$$

(1)

$$b) \frac{dT}{dt} = k(T-S)$$

$$i) \text{ if } T = S + Be^{kt} \text{ is a solution}$$

$$\therefore \frac{dT}{dt} = B \cdot k \cdot e^{kt} \text{ but from } T-S = Be^{kt}$$

$$\therefore \frac{dT}{dt} = k(T-S) \therefore \text{shown}$$

$$ii) t = 0 \quad T = 100^\circ C$$

$$t = 30 \text{ mins } T = 80^\circ C$$

$$t = 60 \text{ (further 30)} \therefore T = ?$$

$$T = S + Be^{kt} \quad S = 25^\circ C$$

$$100 = 25 + B \cdot e^0 \therefore B = 75 \quad (1)$$

$$80 = 25 + 75 \cdot e^{k \times 30}$$

$$\frac{11}{15} = e^{30k}$$

$$\ln\left(\frac{11}{15}\right) = 30k$$

$$\therefore k = -0.010338 \quad (1)$$

$$T = 25 + 75 \cdot e^{-0.010338 \times 60}$$

$$T = 65.33^\circ C \quad (1)$$

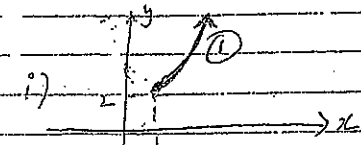
Question 7

13

$$a) f(x) = x^2 - 2x + 3$$

$$\text{axis } \therefore x = \frac{2}{2} = 1$$

$$\therefore \text{vertex } (1, 2)$$



$$i) \text{ largest domain: } x \geq 1 \quad (1)$$

$$ii) f: y = x^2 - 2x + 3$$

$$y - 3 + 1 = x^2 - 2x + 1$$

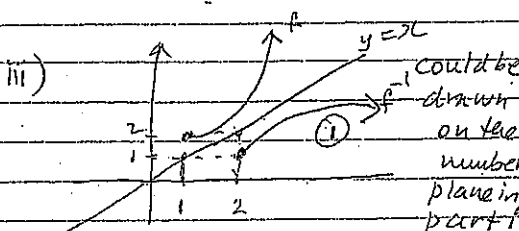
$$y - 2 = (x - 1)^2 \quad (1)$$

$$\sqrt{y - 2} = x - 1$$

$$\therefore x = \sqrt{y - 2} + 1$$

$$\therefore f^{-1}: y = \sqrt{x - 2} + 1 \quad (1)$$

$$\text{domain of } f^{-1}: x \geq 2 \quad (1)$$



$$iv) f(f^{-1}(8)) = 8 \quad (1)$$

$$b) A_1 = 15800 \left(1 + \frac{8.5/2}{100}\right) - M$$

$$(2) A_1 = 15800 \left(1 + \frac{1}{150}\right) - M \quad (1)$$

$$\therefore A_1 = 15800 \times \frac{151}{150} - M$$

$$\therefore A_2 = A_1 \times \frac{151}{150} - M \quad (1)$$

$$= 15800 \times \left(\frac{151}{150}\right)^2 - M \left(\frac{151}{150} + 1\right)$$

$$\therefore A_3 = A_2 \times \frac{151}{150} - M = 15800 \left(\frac{151}{150}\right)^3 - M \left(\frac{151}{150} + 1\right)$$

$$(1) A_n = A_{n-1} \times \frac{151}{150} - M \quad (1)$$

$$= 15800 \times \left(\frac{151}{150}\right)^n - M \left(\frac{151}{150} + \dots + 1\right)$$

$$iii) A_n = 0, M = \$1260$$

$$0 = 15800 \left(\frac{151}{150}\right)^n - 1260 \left[1 + \frac{\left(\frac{151}{150}\right)^n - 1}{\frac{151}{150} - 1}\right] \quad (1)$$

$$0 = 15800 \left(\frac{151}{150}\right)^n - 1260 \times 50 \frac{\left(\frac{151}{150}\right)^n - 1}{\frac{151}{150} - 1}$$

$$0 = 15800 \left(\frac{151}{150}\right)^n - 1260 \times \frac{\left(\frac{151}{150}\right)^n - 1}{\frac{1}{150}}$$

$$1260 \times 150 \times \left[\left(\frac{151}{150}\right)^n - 1\right] = 15800 \left(\frac{151}{150}\right)^n$$

$$189000 \left(\frac{151}{150}\right)^n - 15800 \left(\frac{151}{150}\right)^n = 189000 \quad (1)$$

$$\left(\frac{151}{150}\right)^n [189000 - 15800] = 189000$$

$$\left(\frac{151}{150}\right)^n = \frac{79}{866}$$

$$\ln \left[\left(\frac{151}{150}\right)^n\right] = \ln \frac{79}{866}$$

$$n = \frac{\ln \left(\frac{79}{866}\right)}{\ln \left(\frac{151}{150}\right)} = \frac{\ln 1.0912}{\ln \left(\frac{151}{150}\right)}$$

$$\therefore n = 13.1386 \text{ repayments}$$